

Atomic Force Microscopy - Lab Report

Szegedi Livia, Nguyen Ba Phi

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Abstract

This report details the operational principles and practical applications of Atomic Force Microscopy (AFM) using a Semilab AFM-1000 system. The experiment encompassed system calibration using a TGXYZ-01 grating, the optimization of Proportional-Integral (PI) feedback parameters, and the extraction of critical dimensions from a patterned wafer. Furthermore, surface roughness analysis was performed to compare different semiconductor materials, revealing that the Silicon wafer was exceptionally smooth ($\sigma \approx 0.287 \text{ nm}$) compared to the Gallium-Arsenide sample ($\sigma \approx 3.650 \text{ nm}$). Ultimately, the results demonstrate the high precision of AFM and its critical role in nanoscale process control and material characterization.

Introduction

Atomic Force Microscopy (AFM) is a powerful, high-resolution scanning probe technique that maps surface topography by sensing the interaction forces between a nanoscale tip and a sample. In the semiconductor industry, monitoring surface properties such as critical dimensions and microroughness is vital for ensuring device performance and uniformity. This experiment utilizes the AC (tapping) mode to minimize tip-sample degradation while extracting precise topographical data. The primary objective of this laboratory exercise is to practically explore the AFM workflow—including spatial calibration, feedback loop optimization, and digital signal processing—to accurately characterize the nanoscale geometry and roughness of various semiconductor wafers.

Measurement and Analysis

Task 2: Frequency Sweep

Experimental Setup and Procedure

The objective of this task was to determine the resonance frequency of the cantilever and observe the effect of environmental factors on the optical detector. The measurement was conducted using the Semilab AFM-1000 and the SDF software. Upon navigating to the Frequency Sweep tab, a full sweep was performed from 20 kHz to 500 kHz. To isolate the impact of ambient interference, the full frequency

sweep was conducted twice: first with the top of the AFM machine open, and then again after carefully closing the machine top.

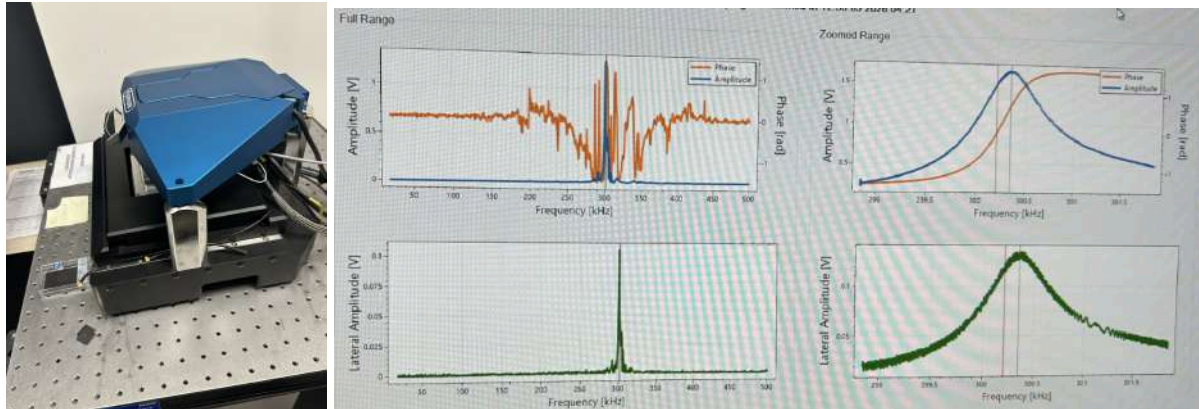


Figure 2a: Image of the setup and figure results in closed state

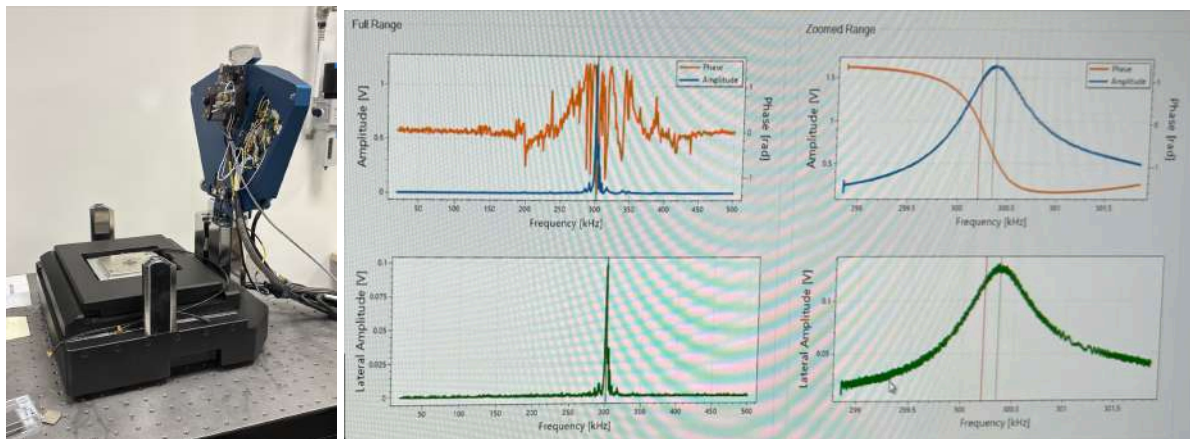


Figure 2b: Image of setup and figure results in open state

Experimental Results

The sweeps successfully identified the cantilever's resonance peak at approximately 300.3 kHz. The extracted metrics were highly consistent between the two states, yielding Quality Factors (Q) of roughly 264.89 and 263.89, with corresponding measured amplitudes of 1.506 V and 1.499 V. However, a major visual difference was observed when comparing the phase spectra. In the open state, the phase response was shifted in reverse by approximately 180 degrees compared to the measurement taken in the closed state.

Data Analysis

The pronounced 180-degree phase shift can be explained by examining the AFM's cantilever detection system. The microscope senses cantilever deflection by aiming a laser at the back of the cantilever and capturing the reflected beam with a photodiode. When the top of the AFM is left open, the optical path of this readout laser is uncovered. As a result, ambient laboratory illumination (such as overhead lights or computer screens) acts as an external light source that generates an artificial signal in the optical detector. This environmental light floods the photodiode, interfering with the intended laser signal. The electronics

misinterpret this high baseline of stray light, resulting in the inverted phase shift. Closing the top shields the detector from ambient light, restoring the true phase and amplitude response

Task 3: XYZ Calibration of XYZ piezo actuators

Experimental Setup and Procedure

The goal of this task was to perform lateral and vertical calibration of the AFM piezo actuators using a standard reference sample. A TGXYZ-01 calibration grating was loaded onto the motorized stage. The AFM was configured in AC (tapping) mode with a scan width of 22.5 μm , 128 points, a scan rate of 1 Hz, and a 30% damping setpoint. After engaging the surface, a preliminary scan was captured. The raw data was then exported to the analysis software to run through a series of correction scripts (Line order 1, LineCorr, Segmentation, XYCalibration, and ZCalibration).

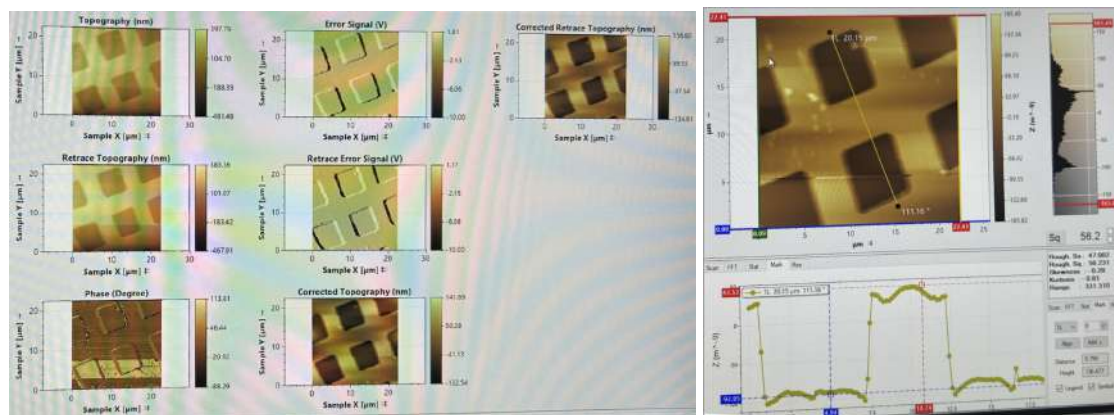


Figure 3a: Topography and images from scan and analysis of corrected retrace topography

Experimental Results

During the setup, the 30% damping setpoint was used, which means that the controller keeps the cantilever oscillation reduced by 30% from its free amplitude. The resulting scan roughly matched the larger pattern visible on the camera image. Some slow, low-frequency height changes were visible in the initial raw line traces, but these were expected as they represent sample tilt and scanner drift rather than real surface structures.

During the initial data processing, the raw histogram showed a significant number of messy peaks. After applying the "Line of order 1" correction, the number of peaks was successfully reduced to exactly two, which is the ideal result representing the top and bottom levels of the grating. When the subsequent "LineCorr" step was added, no further significant changes were observed in the image or histogram. The primary difference between these two operations is that the first corrects the overall macroscopic tilt of the sample, while the second adjusts individual scan lines that might have shifted during the measurement.

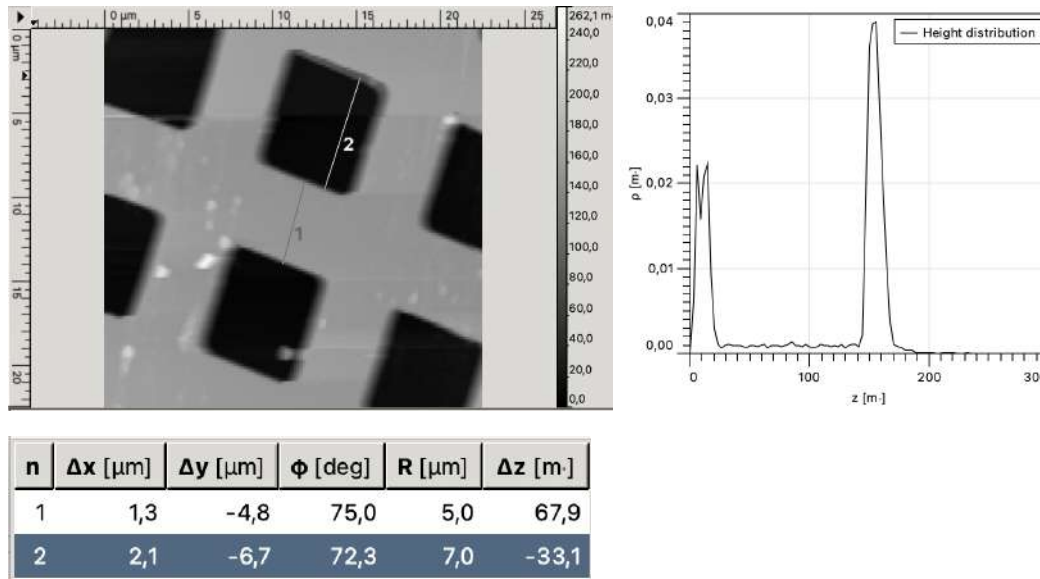


Figure 3b: Digital analysis of topography with calibration with width, height data

Data Analysis and Digital Calibration

To extract highly accurate geometric dimensions, further digital calibrations were performed on the data (represented by the black and white topography images). This deeper post-processing sequence included: leveling the data by mean plane subtraction, correcting horizontal scars, aligning rows by the median of differences, and finally shifting the minimum data point to zero. This fully corrected data clearly indicates that the square features on the grating are dips rather than pillars. Profile analysis of these dips revealed a step height of 150 nm, a distance between features of 5 μm , and a square side length of 7 μm , resulting in a total pitch size of 12 μm .

Task 4: Effects of changing PI parameters

Experimental Setup and Procedure

To evaluate the responsiveness of the AFM's feedback loop, a continuous scan was performed while dynamically altering the Proportional-Integral (PI) settings. The scan rate was intentionally increased from 1 Hz to 2 Hz; this not only expedited the measurement but also forced the controller to react to surface changes faster, making the effects of the parameter tweaks much more noticeable. The experiment was divided into two phases. First, the Integral parameter (Integrator) was fixed at 10 Hz, and the Proportional parameter (Loop Gain) was systematically increased by 0.1 every 10 scan lines, scaling from its minimum limit of 0.1 up to 1.0. Next, the Loop Gain was fixed at a baseline of 0.2, and the Integrator was increased by 10 Hz every 10 scan lines, sweeping from its lower limit of 1 Hz up to 100 Hz.

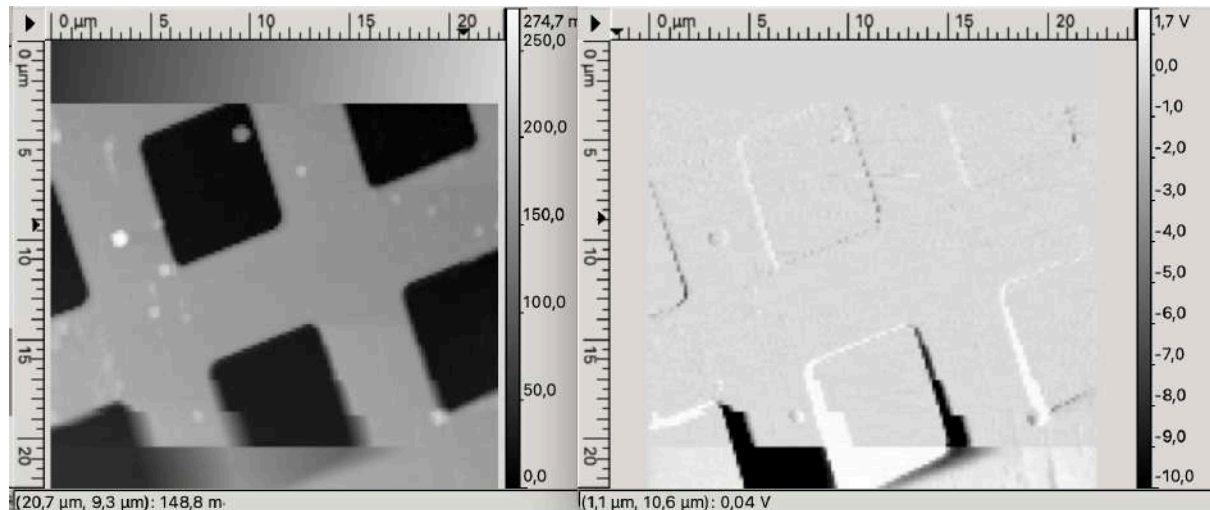


Figure 4a: Topography and error signal with incrementing I from the bottom up

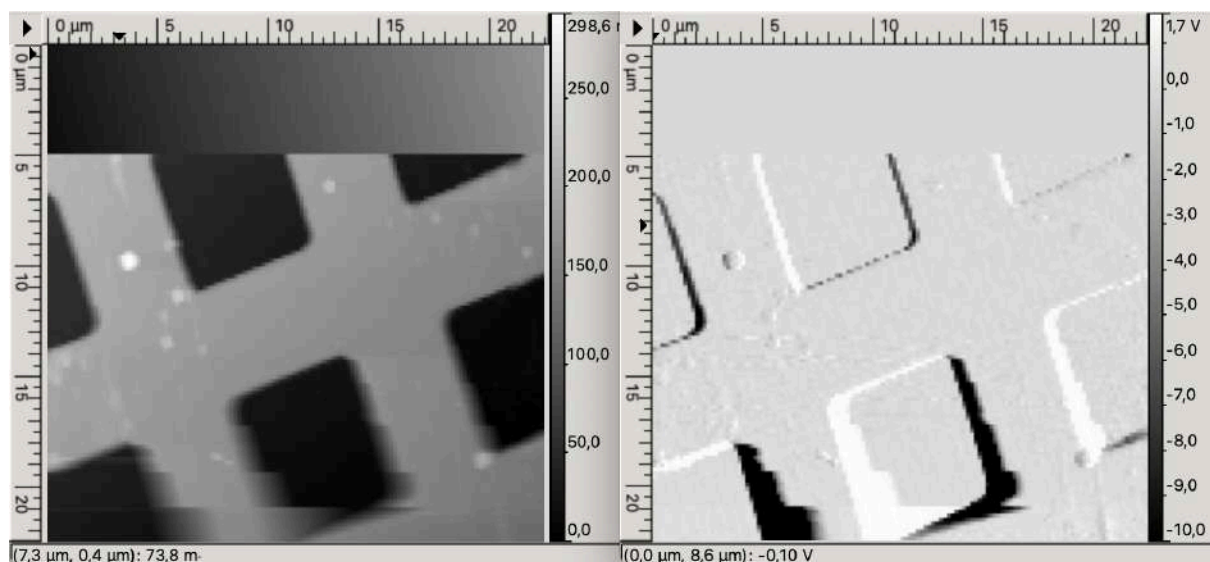


Figure 4b: Topography and error signal with incrementing P from the bottom up

Experimental Results

The systematic alterations in the PI settings manifested as distinct horizontal bands across both the Topography and Error Signal images. When the Loop Gain (P) was set too low, the system exhibited a lazy, slow response, causing the sharp edges of the square features to appear smeared or blurred. Conversely, when P was pushed to its higher limits, the system began to visibly overshoot at the feature edges. Similar behavior was observed during the Integrator (I) sweep: too low of an I value resulted in a sluggish response that struggled to reach the desired height (mimicking the low P behavior), while high I values induced overshooting. Notably, while high integral values are known to cause a system to oscillate wildly, severe continuous oscillation remained practically unnoticeable and negligible within the limits tested here.

Data Analysis

These visual artifacts directly reflect the mathematical roles of the PI controller components. The Proportional (P) term reacts instantaneously to the current error (the immediate difference between the setpoint amplitude and measured amplitude). A low P fails to suppress the error fast enough upon hitting a steep step, causing the "lazy" smearing, while a high P overcompensates. The Integral (I) term, on the other hand, reacts to the accumulated error over time, ensuring the cantilever doesn't just get close, but eventually reaches the exact setpoint. If I is too low, the history of the error isn't factored in fast enough. If I is too high, the accumulated error forces the z-piezo past the target. The fact that extreme oscillation was negligible suggests that the maximum limit of 100 Hz, combined with the damping of the cantilever, was still just within the margins of baseline stability for this specific setup.

Furthermore, the standard deviation of the error signal serves as a direct quantitative measure of the feedback loop's performance. When the PI parameters are optimally tuned, the z-piezo reacts almost instantly to surface variations, keeping the actual amplitude very close to the setpoint; this results in an error signal with a very low standard deviation. Conversely, when the settings (especially P) are too low, the system's sluggish response causes large, sustained deviations from the amplitude setpoint over every surface feature, significantly increasing the standard deviation of the error signal. This sluggish extreme also reveals the phenomenon where the topography and error signals "switch places." At very low feedback settings, the z-piezo barely moves, resulting in a topography image that appears mostly flat and featureless. Because the height is not being adjusted, the cantilever absorbs the full physical impact of the surface geometry, causing the error signal to perfectly map the sample's physical bumps and dips. In this state, the visual representation of the surface structure is effectively transferred from the topography channel into the error signal channel.

Task 5: Critical dimension measurement

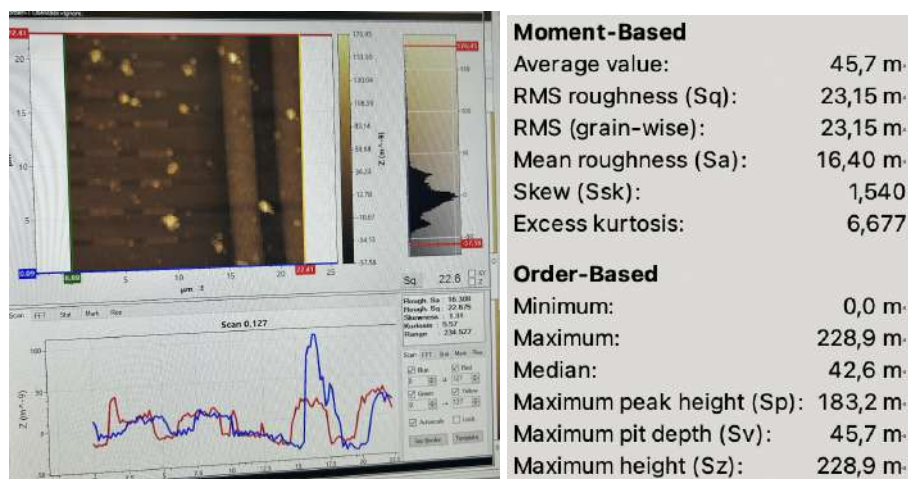


Figure 5a: Corrected topology from measurement with basic descriptive statistics

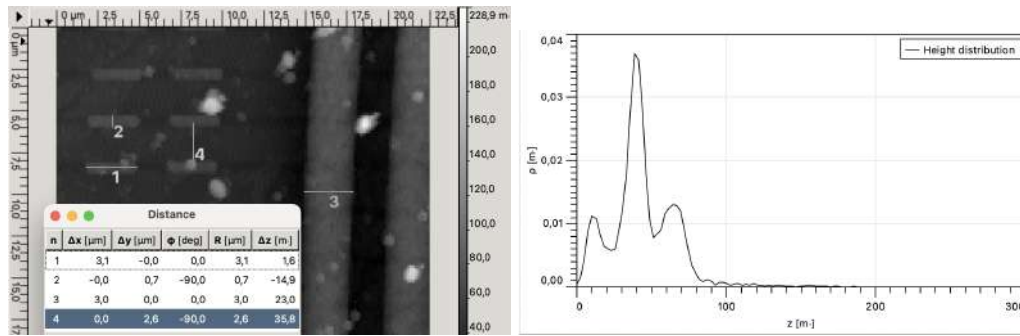


Figure 5b: Digital analysis of topography with measured distance and height distribution

	Width (μm)	Height (10 ⁻⁹ m)	Period (features)
Small horizontal rise	3.1	65	2 per 2.6μm vertically
Large vertical rise	3.0	65	2

Table 5: Table with evaluated width height and period of features

Experimental Setup and Procedure

The objective of this task was to accurately measure the geometric properties (critical dimensions) of a patterned wafer. The sample was loaded into the microscope, and a region exhibiting a distinct periodic pattern was located using the camera. The PI settings were specifically optimized for this new topography to ensure accurate tracking. After capturing the raw scan, the data was exported for digital post-processing to extract the exact widths, heights, and depths of the structural features.

Experimental Results

The scan revealed a complex surface structure featuring large vertical raised lines, smaller horizontal raised segments, and varying background levels (along with a noticeable amount of scattered surface debris). A key observation during the live measurement was that applying a "Line order 1" correction was actually detrimental to the image quality. Because the sample features small, distinct horizontal rises, the line-by-line polynomial fit attempted to incorrectly flatten these valid physical structures into the background. Therefore, accurate dimensional extraction relied on careful digital analysis without this specific live correction.

Data Analysis

Using the digital analysis cross-sections, the critical dimensions of the pattern were successfully quantified. The width of the raised structures is highly consistent; both the small horizontal rises and the large vertical rises measure approximately 3.0 μm across. In terms of vertical dimensions, the small horizontal rises reach the same elevation as the large vertical bands, peaking at roughly 65 nm relative to the lowest measured point on the sample. Furthermore, the profile analysis revealed that the background is not perfectly uniform: the trench (dip) on the right side of the main vertical structure is noticeably deeper

than the region on the left side by approximately 30 nm. The multi-peak height distribution histogram perfectly corroborates these varying base levels and consistent peak heights.

Task 6: Roughness measurement

Experimental Setup and Procedure

The final task aimed to measure and compare the surface roughness of a Silicon (Si) wafer and a Gallium-Arsenide (GaAs) wafer. Each sample was successively loaded into the AFM, and the camera was used to identify a clean, debris-free region. The measurements were performed using a highly localized scan area (1 μm width, 256 points, 1 Hz scan rate) to isolate high-frequency spatial components (microroughness). To establish a baseline for the machine's intrinsic noise, a "zero scan" was also recorded. Following the measurements, standard tilt correction was applied, and the statistical roughness indicators, Mean Roughness (S_a) and RMS Roughness (S_q), were extracted from the error signals.

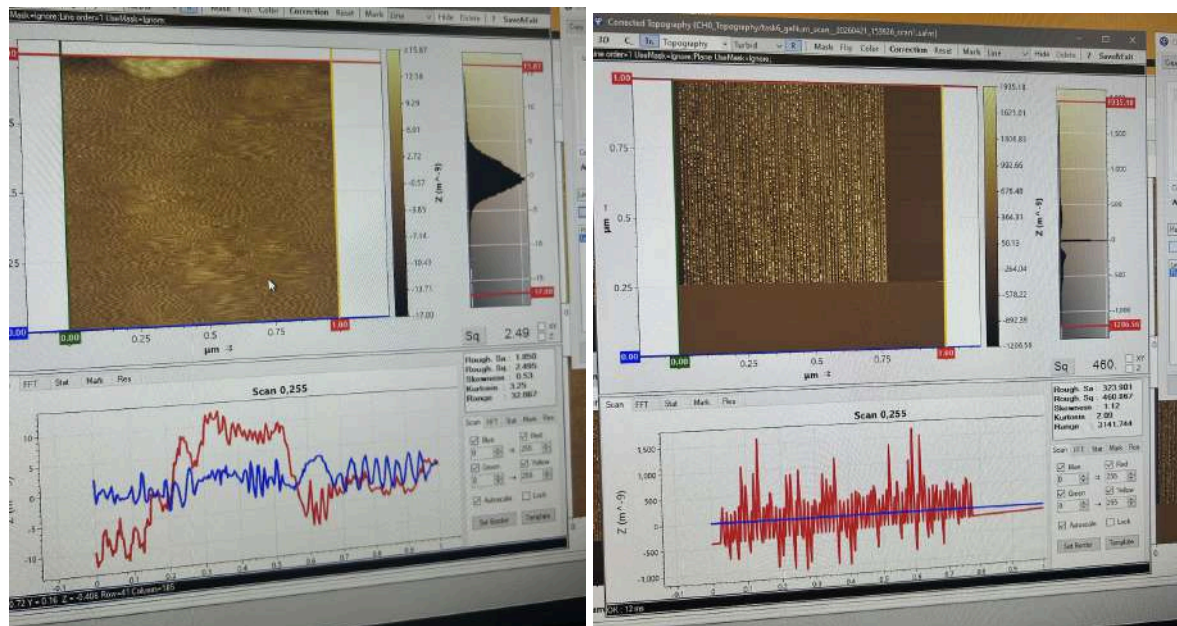


Figure 6a: Some data of silicon (left side) and gallium (right side). Note that there is a major defect/error during the measurement of the gallium scan but was fixed digitally via cropping

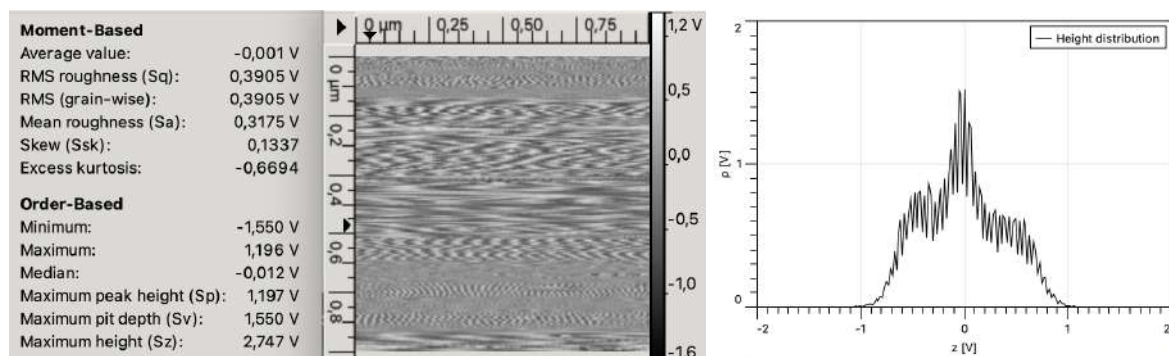


Figure 6b: Digital analysis of error signal of zero scan

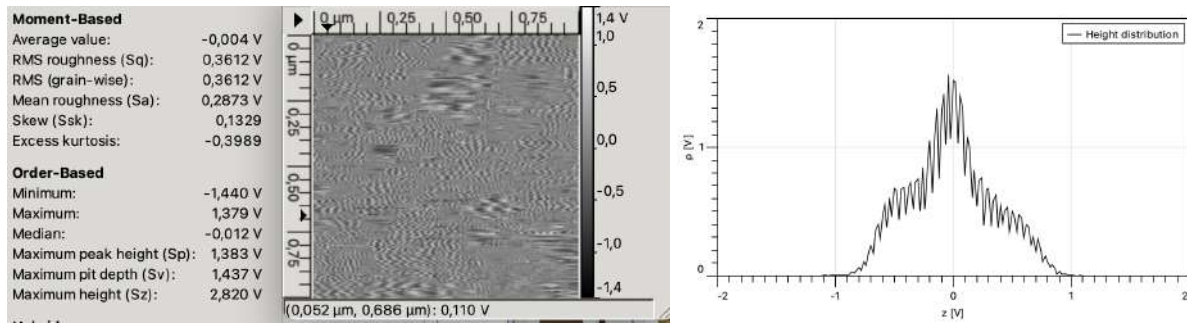


Figure 6c: Images and analysis of error signal silicon scans

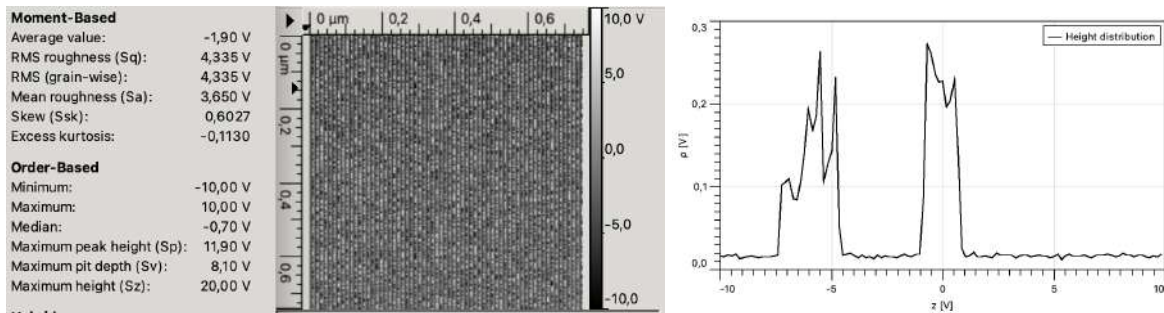


Figure 6d: Digital analysis of error signal of gallium scan

	Average (V)	RMS roughness (V)	Mean roughness (V)	Skew
Zero scan	-0.001	0.3905	0.3175	0.1337
Silicon scan	-0.004	0.3612	0.2873	0.1329
Gallium scan	-1.90	4.335	3.650	0.6027

Table 6: Table of average value, RMS roughness, mean roughness and skew of the scans

Experimental Results

The extracted data revealed a massive difference in surface quality between the two materials. The Gallium-Arsenide sample exhibited a highly rough surface, yielding a Mean Roughness (Sa) of 3.650 V and an RMS Roughness (Sq) of 4.335 V. In stark contrast, the Silicon sample was remarkably smooth, yielding an Sa of 0.2873 V and an Sq of 0.3612 V. Interestingly, the roughness values for the Silicon wafer were actually slightly lower than the baseline zero scan (which had an Sa of 0.3175 V and Sq of 0.3905 V).

Data Analysis

The Silicon sample is clearly much smoother than the Gallium-Arsenide sample, which makes sense given its role in the semiconductor industry. For modern Silicon-based MOS transistors, extremely low surface roughness is a strict requirement. High microroughness at the Si-SiO₂ interface leads to frequent charge

carrier collisions, which increases scattering and severely degrades electron mobility. Therefore, Silicon wafers undergo aggressive and highly mature chemical-mechanical polishing (CMP) and cleaning steps (like RCA cleans) to ensure atomic-level flatness. The slight discrepancy where Silicon measured "less rough" than the zero scan simply indicates that the physical Silicon surface is exceptionally flat, so flat, in fact, that its actual roughness is at or below the intrinsic electronic and mechanical noise floor of the AFM system itself.

Conclusion

In conclusion, this laboratory exercise successfully demonstrated the complete workflow required for accurate AFM-based nanoscale characterization. By performing lateral and vertical calibrations, the system was tuned to accurately extract the critical dimensions of micro-structures on a patterned wafer. Additionally, the exploration of PI parameters highlighted the delicate balance required in feedback loop tuning to prevent scanning artifacts. Finally, the roughness measurements quantitatively verified the atomic-level smoothness of semiconductor-grade Silicon compared to Gallium-Arsenide, underscoring AFM as an indispensable and highly sensitive tool for quality control in semiconductor manufacturing.